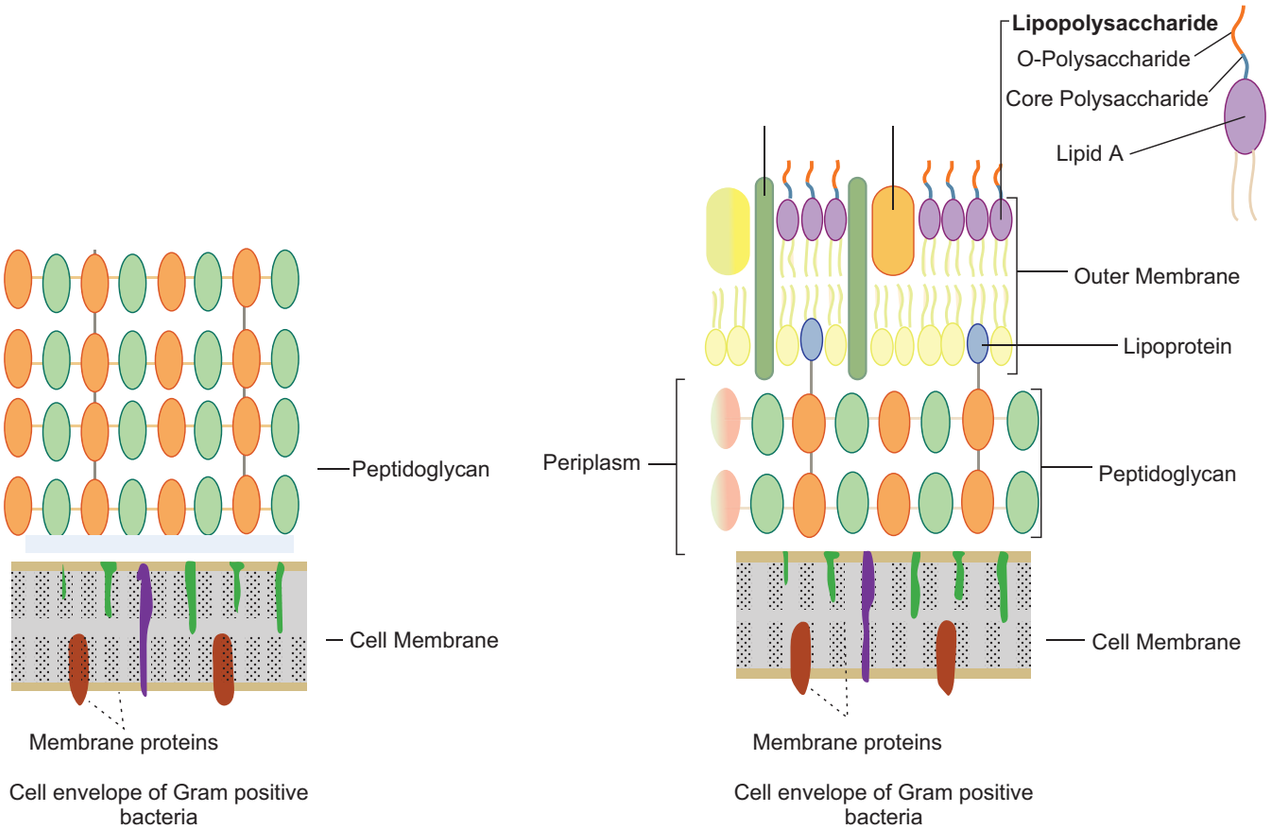


Figure 7.11: Cell envelope of Gram positive and Gram negative bacteria

Cell wall may contain other substances in addition to peptidoglycan. For instance, *Staphylococcus aureus* and *Streptococcus fecalis* contain **teichoic acids** (polymer of acidic polysaccharides) covalently linked to peptidoglycan. Cell wall of Gram positive bacteria contain very little lipid but *Mycobacterium* and *Corynebacterium* cell walls are rich in **mycolic acid** (or Cord factor) which make them acid fast. When stained, the cells cannot be decolorized easily despite treatment with dilute acids. *Mycoplasma* lack cell wall.

Protoplast is a bacterial cell consisting of cell material bound by a cytoplasmic membrane.

Spheroplast is a bacterial cell with two membranes namely the cytoplasmic membrane and the outer membrane but no cell wall.

Functions of cell wall

- It gives shape to bacteria like a bicycle tyre that maintains the necessary shape and prevents the more delicate inner tube (the cytoplasmic membrane) from bursting when it is expanded.
- It protects bacteria from osmotic lysis in dilute solutions (hypotonic environment).
- It protects cell from toxic substances.

HOTS

How do bacteria maintain their shape?

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